

Predicted preheater cyclone jamming and kiln ring formation on a 7,000 TPD line, with lead time to the operator

A leading cement manufacturer — 7,000 TPD kiln line losing clinker output and fuel to cyclone build-up

Scope

- ▮ Cover the two-string, six-stage preheater, where most blockages sat in the bottom cyclone (Cyclone 6)
- ▮ Build one solution across cyclone jamming and kiln ring formation identification
- ▮ Give the operator lead time on a developing jam, not an alarm once ΔP has already risen

Approach

- ▮ Integrated process data from DCS and SCADA across both preheater strings — cyclone and riser pressures, stage temperatures, draft, O_2 and kiln load
- ▮ Integrated chemistry data — raw-mix and hot-meal chemistry, with the volatile circulation drivers that precede build-up
- ▮ Integrated actual historical blockage events from the CCR logbook, so the plant's own recorded events define what a real event looks like
- ▮ Built the risk mechanism for Cyclone 6, where the majority of the blockages occurred
- ▮ Output gives the risk at a given point and the lead time to likely jamming, with a confidence score
- ▮ Each alert carries its contributing signals and a recommended action for the operator — advisory, with the kiln team deciding

Outcomes

- ▮ 9-month pre and post study on the same kiln line
- ▮ Before: 13 kiln shutdowns from Cyclone-6 blockage, 46 hours of unplanned downtime
- ▮ After: 3 blockages, 13 hours of unplanned downtime — ~70% reduction
- ▮ 33 hours of kiln downtime recovered, about 9,600 t of clinker at 7,000 TPD
- ▮ Fewer kiln startups also improved specific heat consumption

Predicted clinker C3S ahead of the lab result and used the trend to guide alternative fuel rate optimisation

A leading cement manufacturer seeking earlier sight of clinker quality and higher TSR

Scope

- ▮ Predict clinker C3S ahead of the lab result
- ▮ Use the predicted trend to guide the operator on TSR (alternative fuel rate) optimisation
- ▮ Reduce clinker quality variability rather than correcting it after the lab reports

Approach

- ▮ Integrated plant data across process, fuel and quality systems — full input list [TBC, Vivek]
- ▮ C3S predicted at every 15-minute interval, in place of waiting for the lab test
- ▮ Prediction surfaced live to the control room, so quality is visible within the shift
- ▮ C3S trend used to drive operator recommendations on the TSR side
- ▮ Model tuned to the plant's own raw mix and fuel behaviour
- ▮ Recommendations kept advisory — the operator decides the fuel change

Outcomes

- ▮ Standard deviation in C3S reduced — pre [TBC] to post [TBC]
- ▮ Lower process variability on clinker quality
- ▮ Clinker quality visible every 15 minutes instead of at lab frequency
- ▮ TSR increase of approximately 1.5–2 percentage points (indicative)

Moved 600 rotating assets across four power plants from manual checks to daily condition-based predictive maintenance

A leading thermal power producer checking rotating equipment manually at four plants

Scope

- ▮ Cover 600 assets across four plants with condition-based predictive maintenance for rotating equipment
- ▮ Move from manual checks to a daily health view of every covered asset
- ▮ Give the maintenance team the root cause and the action, not just an alarm

Approach

- ▮ Deployed triaxial vibration sensors on the rotating equipment across the four plants
- ▮ Measured at every point: vibration in all three directions, noise and temperature
- ▮ Model computes a severity and health score for each asset on a daily basis
- ▮ Assets bifurcated into low, medium, high and highest risk categories
- ▮ Risk, root cause and recommendation issued at equipment level to stop unplanned downtime
- ▮ Faults covered are mainly bearing faults and mechanical wear — structural looseness, misalignment and rotating looseness

Outcomes

- ▮ Unplanned downtime reduced 75%
- ▮ Maintenance cost reduced 20%
- ▮ 1,500 man-days saved in the first 6 months
- ▮ ROI realised in under 1 year

Optimised captive power plant operations with sootblower automation, coal-feeder control and AI-agent diagnosis

An integrated pulp and paper mill's captive power plant seeking steadier boiler operation

Scope

- ▮ Cut the auxiliary steam spent on boiler cleaning without losing heat-transfer efficiency
- ▮ Reduce coal-feed variability and hold bed stability on the AFBC boiler
- ▮ Shorten diagnosis of boiler, turbine and generator problems across both process and maintenance causes

Approach

- ▮ Boiler inputs: flue-gas temperatures across passes, draft and pressure profile, steam flow and temperature, feed-water and attemperation, and soot-blower operating history
- ▮ Cleaning sequenced on measured fouling rather than a fixed clock — 29 soot blowers on the 16 MW recovery boiler, in closed loop
- ▮ Coal-feed inputs: feeder speed and load, bed temperature and pressure profile, primary and secondary air, and fuel quality variation — 7 feeders on the 21 MW AFBC boiler (in progress)
- ▮ Diagnostic inputs: DCS process tags and lab results across boiler, turbine, generator and auxiliaries on the 21 MW cogeneration plant
- ▮ Diagnostic logic co-developed with a domain SME of 40+ years, so the plant's own operating know-how sits behind each recommendation
- ▮ Every recommendation stays advisory — the operator approves the action, and outcomes are reviewed to refine the guidance

Outcomes

- ▮ Auxiliary steam saved: 20 TPD, worth ₹1 Cr a year
- ▮ Wash cycles and tube erosion reduced — longer element life
- ▮ Flue-gas heat recovery improved from cleaner surfaces
- ▮ Coal-feed variability and bed stability — AFBC boiler control in progress

Replaced manual day-ahead and intraday power balancing across a metals group's plants with slot-level optimisation

A leading metals group balancing power across its plants within a state manually, day-ahead and intraday

Scope

- ▮ Optimise power allocation across the group's plants within the state, day-ahead and intraday
- ▮ Bring generation, consumption, exchange purchases and renewable sources into one balance
- ▮ Minimise the losses that come from energy being mis-allocated at slot level

Approach

- ▮ Day-ahead generation and consumption data ingested for all 96 slots
- ▮ Intraday changes picked up on both the generation and consumption side, including unplanned downtime
- ▮ Sources covered across captive generation, grid, power exchange and solar, including external solar
- ▮ CTU and STU compliances built into the model and held in configuration mode, so a new utility rule is a setting rather than a rebuild
- ▮ Slot-level allocation optimised in place of the manual day-ahead and intraday exercise
- ▮ Addresses the two to three slots (30–45 minutes) typically lost to manual reporting, where unbalanced energy is given free to the grid or bought back at a premium

Outcomes

- ▮ Manual power-balancing effort eliminated
- ▮ Reporting lag to the SLDC cut from 30–45 minutes to about a minute
- ▮ Two to three slots per cycle recovered that were previously lost to manual reporting
- ▮ Potential saving of around ₹6 Cr a year